Practice Session

Question-1

The **Lahore Airport Authority** wants every Aircraft object created in the system to share a single counter that tracks the total number of aircraft registered across the whole fleet. **Suggest a solution.**

```
class Aircraft {
};
```

Question-2

Flight class has a static data member static int activeFlights;. From main(), without creating any Flight object, you want to print the current number of active flights. **How will you do that**?

```
class Flight {
    static int activeFlights;
public:
    // some function here
};
int Flight::activeFlights = 0;
```

Question-3

What is missing here?

```
class Flight {
    int number;
    float cost;
};
int main() {
    int num = 1;
    Flight flight(num);
}
```

Question-4

What is missing here?

```
class Flight {
    int number;
    float cost;
};
int main() {
    int num = 1;
    float cost = 100f;
    Flight flight(num, cost);}
```

Question-5

What is missing here?

```
class Flight {
    int number;
    float cost;
};
int main() {
    int num = 1;
    Flight flight = num;
}
```

Question-6

The Flight class stores string origin; and string destination;. Write a member function **Bool isSameRoute(Flight other);** that returns true only if this and the passed-in Flight object share both the same origin and the same destination.

```
class Flight {
    string origin;
    string destination;
};
```

Question-7

Output?

```
class Runway {
    int length;
     public:
    Runway(int l) : length(l) {}
    Runway extend(int extra) {
        return Runway(length + extra);}
    int getLength() { return length; }
};
int main() {
    Runway r1(3000);
     ________= r1.extend(500);
    cout << r2.getLength();}
```

Question-8

You have two classes: Passenger and BoardingPass.

Write a member function inside Passenger:

BoardingPass issueBoardingPass(int seatNum, string flightCode);

that constructs and returns a BoardingPass object for this passenger.

```
class BoardingPass {
public:
    string passengerName;
    int seatNumber;
    string flightCode;
    BoardingPass(string n, int s, string f)
        : passengerName(n), seatNumber(s){};
};
class Passenger {
    string name;
public:
    Passenger(string n) : name(n) {}
    // Write the function below};
```

Question-9

Which of the following correctly accesses the getModel() member of plane?

```
Aircraft* plane = new Aircraft("A380");
```

A. plane.getModel();

B. plane->getModel();

C. (*plane).getModel();

D. Both B and C